NORDHAVEN

Urban Infrastructure Mapping & Geospatial Data Engineering

1. Project Overview

Project type: Independent geospatial infrastructure mapping case study
Study area: Bremen, Germany
Coordinate reference system: ETRS89 / UTM Zone 32N (EPSG:25832)
Primary GIS platform: QGIS 3.10.13
Database format: GeoPackage

Project Objective

The objective of this project is to develop a structured geospatial infrastructure database by integrating reference datasets, spatial digitisation, infrastructure classification, attribute management and quality assurance procedures.

The project demonstrates a practical geospatial data production workflow, including the creation and organisation of infrastructure layers, management of spatial attributes, geometry validation and cartographic presentation.

Project Scope

The demonstration covers:

- Road and paved-area mapping
- Infrastructure point features
- Secondary infrastructure line features
- Traffic sign representation and classification
- Attribute population and management
- Spatial geometry validation
- Map composition and project documentation

Project Status and Data Disclaimer

NORDHAVEN is an independently developed portfolio case study based on professional experience in geospatial infrastructure data production. Its demonstration datasets, schemas, classifications and outputs are independently assembled and are not client deliverables or reproductions of confidential employer data.

2. Data Sources

Primary reference data: OpenStreetMap (OSM), Bremen, Germany.

The project uses OSM-derived vector reference layers for roads, buildings and land-use features. These datasets provide the base spatial context for the infrastructure mapping workflow.

Data attribution: © OpenStreetMap contributors
Data licence: Open Database License (ODbL) 1.0
Licence information: https://www.openstreetmap.org/copyright

The exact download provider and extraction date are yet to be verified.

3. Data Preparation and GIS Workflow

3.1 Project Environment and Spatial Reference

The project was organised using QGIS 3.10.13, with a structured GeoPackage-based data management approach. The working coordinate reference system was set to ETRS89 / UTM Zone 32N (EPSG:25832), providing a projected coordinate framework suitable for spatial measurement and infrastructure mapping in the Bremen study area.

3.2 Reference Data Preparation

OpenStreetMap-derived vector datasets representing roads, buildings and land-use features were loaded into the QGIS project. These datasets provided the underlying spatial context for infrastructure interpretation and digitisation.

Raster imagery, where available, was used as a visual reference for identifying infrastructure boundaries, asset locations and surface characteristics.

3.3 Infrastructure Feature Digitisation

Separate GIS layers were organised according to geometry type and infrastructure category.

- **Polygon features:** Road areas, parking areas, pedestrian areas and other surface categories.
- **Point features:** Selected infrastructure assets such as manholes, streetlights and trees.
- **Line features:** Linear infrastructure such as curbs, drainage channels and crash barriers.
- **Traffic sign features:** Independently created demonstration features representing regulatory, warning and information signs.

Features were digitised and organised into their respective layers using the available spatial references.

3.4 Attribute Management

A project-specific attribute structure was designed to record feature identifiers, infrastructure classifications, surface materials, source references and review status.

Attribute values were populated using the independently defined demonstration schema. This provides a consistent structure for storing and reviewing the mapped infrastructure features.

3.5 Cartographic Output

A QGIS print layout was prepared to communicate the mapped infrastructure layers. The layout includes a map frame, legend, north arrow, scale bar and project title. The exported map serves as the primary visual output of the case study.

4. Quality Assurance and Quality Control (QA/QC)

Quality assurance activities were incorporated into the GIS workflow to improve the consistency and reliability of the mapped infrastructure dataset.

4.1 Geometry Validation

QGIS geometry validation was performed on the selected infrastructure layers using the Check Validity tool. The tested layers returned zero invalid geometries, indicating that no geometry errors were identified during this validation.

4.2 Attribute Completeness

An attribute completeness check was performed on the road areas layer to identify records with missing values in the key fields:

- Area classification
- Surface material

- Area measurement
- Source reference
- Review status

The query returned zero selected features, indicating that no missing values were identified in these fields at the time of checking.

4.3 Visual Review

The mapped features were reviewed visually within the QGIS project to assess their representation and consistency with the available reference data.

4.4 QA/QC Scope

The checks documented above represent the validation activities completed for this portfolio case study. They should not be interpreted as a complete assessment of positional accuracy, thematic accuracy or compliance with any external production specification.

5. Results and Deliverables

5.1 Geospatial Database

A structured GeoPackage database was created to organise the independently developed infrastructure layers. The database provides a consolidated spatial data environment for storing and managing the mapped features and their associated attributes.

5.2 QGIS Project

A dedicated QGIS project was configured to organise the reference datasets, infrastructure layers, symbology and map presentation. The project provides a reusable environment for viewing and managing the case study data.

5.3 Infrastructure Overview Map

A cartographic layout was developed to present the mapped infrastructure features. It includes:

- Road and surface area classifications
- Infrastructure point features
- Linear infrastructure features
- Traffic sign demonstration features
- Map legend, north arrow and scale bar

5.4 Deliverable Summary

The principal outputs of the case study are:

1. **Geospatial database:** Nordhaven_Infrastructure.gpkg
2. **QGIS project:** Nordhaven_Master.qgz
3. **Map output:** Infrastructure Overview Map (PDF)
4. **Technical documentation:** Project methodology and QA/QC report

These deliverables demonstrate a structured approach to geospatial data organisation, infrastructure mapping, attribute management and cartographic presentation.

6. Limitations and Future Improvements

6.1 Current Project Limitations

The NORDHAVEN case study was developed as an independent portfolio exercise. Its scope is limited to the datasets, infrastructure features and validation activities included in the demonstration project.

The following limitations apply:

- OpenStreetMap reference data may not contain every infrastructure asset or provide the level of detail required for detailed engineering applications.
- The mapped infrastructure features are demonstration data and should not be treated as an authoritative municipal asset inventory.
- Geometry and attribute validation was limited to the checks documented in this report.
- Advanced positional accuracy assessment and automated thematic validation have not yet been implemented.

6.2 Future Improvements

The project can be extended through the following technical enhancements:

1. **Python and GeoPandas:** Automate repetitive geospatial processing and attribute validation tasks.
2. **PostGIS integration:** Introduce a spatial database environment for scalable data storage and querying.
3. **Advanced QA/QC:** Develop additional checks for duplicate features, spatial overlaps, gaps and classification consistency.
4. **Remote sensing integration:** Incorporate additional aerial imagery and raster-based analysis.

5. **Automated reporting:** Generate validation summaries and structured quality reports.

6.3 Conclusion

NORDHAVEN demonstrates an end-to-end geospatial infrastructure mapping workflow, covering spatial data preparation, feature digitisation, attribute management, basic quality validation and cartographic output.

The case study provides a foundation for further development in geospatial data engineering, infrastructure mapping and spatial data quality management.